12. Equations in One Variable

Practice Set 12.1

1. Each equation is followed by the values of the variable. Decide whether these values are the solution of that equation.

① $x-4=3$, $\boxed{x=-1, \boxed{7}, -7}$

→ $x=-1$

$-1-4=3$

$-5 \neq 3$

∴ -1 is not solution.

$x=7$

$7-4=3$

$3=3$

∴ 3 is the solution

$x=-7$

$-7-4=3$

$-11 \neq 3$

∴ -7 is the solution.

~~② 9a + 4 = 0   a = 2, 2, 1~~

② $9m=81$, $\boxed{m=3, 9, -3}$

$= \frac{\cancel{9}m}{\cancel{9}} = \frac{\cancel{81}^{9}}{\cancel{9}}$

$\boxed{m=9}$

③ $3-y=4$, $\boxed{y=\boxed{-1}, 4, 2}$

$= y = 4 \neq 3$

$= \boxed{y=-1}$

③ $2a+4=0$, $\boxed{a=2, \boxed{-2}, 1}$

→ $2a=-4$

$a = \frac{-\cancel{4}^{2}}{\cancel{2}}$

$\boxed{a=-2}$

∴ $a=2$ is not solution.

$a=-2$ is the solution

$a=1$ is the solution

2. Solve the following equations.

① $17p - 2 = 49$

$\rightarrow 17p = 49 + 2$

$17p = 51$

$p = \frac{\cancel{51}^{3}}{\cancel{17}_{1}}$

$\boxed{p = 3}$

④ $5(x-3) = 3(x+2)$

$\rightarrow 5x - 15 = 3x + 6$

$5x - 3x = 15 + 6$

$2x = 21$

$\boxed{x = \frac{21}{2}}$

⑥ $\frac{y}{7} + \frac{y-4}{3} = 2$

$\rightarrow \therefore \frac{3y + 7(y-4)}{7 \times 3} = \frac{2}{1}$

$\therefore \frac{3y + 7y - 28}{21} = \frac{2}{1}$

$\therefore \frac{10y - 28}{21} = \frac{2}{1}$

$10y - 28 = 21 \times 2$

$10y - 28 = 42$

$10y = 42 + 28$

$10y = 70$

$y = \frac{\cancel{70}^{7}}{\cancel{10}_{1}}$

$\boxed{y = 7}$

② $2m + 7 = 9$

$\therefore 2m = 9 - 7$

$\therefore 2m = 2$

$m = \frac{\cancel{2}^{1}}{\cancel{2}}$

$\boxed{m = 1}$

⑤ $\frac{9x}{8} + 1 = 10$

$\therefore \frac{9x}{8} + \cancel{1} - \cancel{1} = 10 - 1$

$= 9$

$\frac{9x}{8} =$

$\frac{9x}{\cancel{8}} \times \cancel{8} = 9 \times 8$

$9x = 72$

$\frac{\cancel{9}x}{\cancel{9}} = \frac{\cancel{72}^{8}}{\cancel{9}}$

$\boxed{x = 8}$

③ $3x + 12 = 2x - 4$

$3x - 2x = -4 - 12$

$x = -16$

$\boxed{x = -16}$

(7) $13x - 5 = \frac{3}{2}$

$\rightarrow \quad 13x = \frac{3}{2} + 5$

$= \frac{10+3}{2}$

$13x = \frac{13}{2}$

$\frac{\cancel{13}x}{\cancel{13}} = \frac{\frac{\cancel{13}}{2}}{\cancel{13}}$

$\boxed{x = 2}$

(8) $3(y+8) = 10(y-4) + 8$

$\rightarrow \quad \therefore 3y + 24 = 10y - 40 + 8$

$3y + 24 = 10y - 32$

$3y + 10y = 32 - 24$

$7y = 56$

$y = \frac{\cancel{56}\,8}{\cancel{7}\,1}$

$\boxed{y = 8}$

(10) $\frac{y-4}{3} + 3y = 4$

$\rightarrow \quad \therefore \frac{(y-4)}{\cancel{3}} \times \cancel{3} + 3y \times 3 = 4 \times 3$

$(y-4) - 9y = 12$

$y - 9y = 12 + 4$

$10y = 16$

$y = \frac{\cancel{16}\,8}{\cancel{10}\,5}$

$\boxed{y = \frac{8}{5}}$

(9) $\frac{x-9}{x-5} = \frac{5}{7}$

$\rightarrow \quad \therefore (x-9)7 = (x-5)5$

$7x - 63 = 5x - 25$

$\therefore 7x - 5x = 63 - 25$

$2x = 38$

$x = \frac{\cancel{38}\,19}{\cancel{2}\,1}$

$\boxed{x = 19}$

(11) $\frac{b + (b+1) + (b+2)}{4} = 21$

$\rightarrow \frac{b + (b+1) + (b+2)}{\cancel{4}} \times \cancel{4} = 21 \times 4$

$= 84$

$3b + 3 = 84$

$3b = 84 - 3$

$3b = 81$

$b = \frac{\cancel{81}\,27}{\cancel{3}\,1}$

$\boxed{b = 27}$

(10) $\frac{y-4}{3} + 3y = 4$

$\rightarrow \frac{y-4}{3} + \frac{3y}{1} = \frac{4}{1}$

$\frac{y-4+9y}{3\times1} = \frac{4}{1}$

$\frac{y+9y-4}{3} = \frac{4}{1}$

$\frac{10y-4}{3} = \frac{4}{1}$

$10y - 4 = 3\times4$

$10y - 4 = 12$

$10y = 12+4$

$10y = 16$

$y = \frac{\cancel{16}\,8}{\cancel{10}\,5}$

$\boxed{y = \frac{8}{5}}$

(11) $\frac{b+(b+1)+(b+2)}{4} = 21$

$\rightarrow b+b+1+b+2 = 4\times21$

$3b+3 = 84$

$3b = 84-3$

$3b = 81$

$b = \frac{\cancel{81}\,27}{\cancel{3}}$

$\boxed{b=27}$

[Word problem]

Practice Set 12.2

1. Mother is 25 year older than her son. Find son's age if after 8 years ratio of son's age to mother's age will be $\frac{4}{9}$.

$\rightarrow$ Suppose,

Present age of son is x year.

$\therefore$ Present age of mother is $x+25$ year

After 8 year age of son $= x+8$ year

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After 8 year age mother $= x+25+8$
$= x+33$ year

$\therefore$ By Cond^n,

$$\frac{x+8}{x+33} = \frac{4}{9}$$

$$9(x+8) = 4(x+33)$$
$$9x+72 = 4x+132$$
$$9x-4x = 132-72$$
$$5x = 60$$
$$x = \frac{60}{5}$$
$$\boxed{x = 12}$$

$\therefore$ Son's age is 12 year.

⑦ Of the three consecutive natural numbers, five times the smallest number is 9 more than four times the greatest number, find the number.

→ Three consecutive natural number x, $x+1$, $x+2$

5 time smallest number $= 5x$

4 time greatest $= 4(x+2)$

5 is greater than $4(x+2)$ by 9

$\therefore 5x = 4(x+2)+9$

$\therefore 5x = 4x+8+9$

$\therefore 5x = 4x+17$

$\therefore 5x-4x = 17$

$\therefore \boxed{x=17} \leftarrow \boxed{x}$

$(x+1) = 17+1 = \boxed{18 = x+1}$

$(x+2) = 17+2 = \boxed{x+2 = 19}$

6* Some tickets of ₹ 200 and some of ₹ 100, of a drama in a theaters were sold. The number of tickets of ₹ 200 sold was 20 more than the numbes of tickets of ₹ 100. The total amount received by the theatere by sale of tickets was ₹ 37000. Find the number of ₹ 100 tickets sold.

→ Suppose, Tickets of ₹ 100 be x.

Tickets of ₹ 200 will be $x+20$

Amount of ₹ 100 tickets $= 100x$

Amount of ₹ 200 tickets $= 200(x+20)$

Total amount received = 37000 ₹.

$$100x + 200x + 4000 = 37000$$

$$300x + 4000 = 37000$$

$$300x = 37000 - 4000$$

$$300x = 33000$$

$$x = \frac{330\cancel{00}}{3\cancel{00}}$$

$$\boxed{x = 110}$$

No. of tickets of ₹ 100 is 110.

2. The denominater of a fraction is greater than its numerator by 2. If the numerator is decreased by 2 and denominator is increased by 7, the the new fraction is equivalent with $\frac{1}{2}$. Find the fraction.

→ Numerator of fraction is x

The denominator greater is 12

$\therefore \frac{x}{x+12}$

Numerator decreased by 2

$x-2$

dinominator increased by 7

$x+7$

$\therefore x+12+7 = x+19$

$\therefore$ New fraction is $\frac{x-2}{x+19} = \frac{1}{2}$

$\therefore 2x-4 = x+19$

$2x-x = 19+4$

$x = 23$

Fraction will be $\frac{23}{35}$

3. The ratio of weights of copper and zinc in brass is 13:7. Find the weight of zinc in a brass untensil weighting 700 gram.

→ Ratio of copper and zinc weight is 13:7

$\therefore$ Weight of copper is $13x$ g.

Weight of zinc is $7x$ g.

$13x+7x = 700$

$\frac{\cancel{20x}}{\cancel{20}} = \frac{\cancel{700}}{\cancel{20}}$ 35

$\boxed{x = 35}$

Weight of copper is $13x = 13x \times 35 = 455$ g

Weight of zinc is $7x = 7x \times 35 = 245$ g

The weight of untensils is 245

$\frac{\cancel{7x}}{\cancel{7}} = \frac{\cancel{245}}{\cancel{7}}$ 35

$\boxed{x = 245}$

4* Find three consecutive whole numbers whose sum is more than 45 but less than 54.

→ Let the three conescutive number be, $x, x+1, x+2$

Now. $x+x+1+x+2>45$

$3x+3>45$

substract 3 on both side

$3x+\cancel{3}-\cancel{3}>45-3$

$\cancel{\frac{3x}{3}}>\cancel{\frac{42}{3}}$ 14

$\boxed{x>14}$

$\boxed{x=15}$

$\boxed{x=16}$

$\boxed{x=17}$

OR

$x+x+1+x+2<54$

$3x+3<54$

Substraction 3 on both side

$3x+\cancel{3}-\cancel{3}<54-3$

$\frac{3x}{3}<\cancel{\frac{54}{3}}$ 17

$\boxed{x=16}$

$\boxed{x=17}$

$\boxed{x=18}$

$\boxed{x=17}$

~~x=18~~

~~x~~

5. In a two digits number, digit at the ten's place is twice the digit at unit's place. If the number obtained by interchanging the digits is added to the original number, the sum is 66. Find the number.

→ Suppose, The digit at unit place be x.

The digit at tens place is twice at the unit place

The numbers $= 10\times 2x + x\times 1$

$= 20+x$

x at tens place & $2x$ at unit place

∴ The new no. $= 10\times 2x + x\times 1$

$= 12x = \cancel{12x}\ 10x+2x$

From the given condition

$21x+12x=66$

$33x=66$

Divide by 33 on both side.

$$\frac{\cancel{33}x}{\cancel{33}} = \frac{\cancel{66}^{2}}{\cancel{33}_{1}}$$

$\boxed{x=2}$

The number is $21x = 21 \times 2$
$= 42$

The original no. is 42

8. Raju sold a bicycle to Amit at 8% profit. Amit repaired it spending ₹ 54 Then he sold the bicycle to Nikhil for ₹ 1134 with no loss and no profit. Find the cost price of the bicycle for which Raju purchased it.

→ Suppose the bicycle that Raju purchesed be x

Raju sold his bicycle to Amit al 8% profit.

8% profit on ₹ x

$$\frac{x \times \cancel{8}^{4}}{\cancel{100}_{50}}$$

$$= x \times \frac{\cancel{4}}{\cancel{50}}$$

8% profit on ₹ $\frac{2x}{25} = x \times \frac{2}{25}$

Raju sold the bicycle to Amit for Rs

$$\boxed{x + \frac{2x}{25}} = \frac{25x + 2x}{25} = \frac{27x}{25}$$

Amit spend 54 ₹ for repairing cost of bicycle for Amit is ₹ $\frac{27x}{25} + 54$

Amit sold the bicycle to Nikhil for ₹ 1134 with number of loss and number of profit

$\therefore \frac{27x}{25} + 54 = 1134$

Subtract 54 on both side

$\therefore \frac{27x}{25} + 54 - 54 = 1134 - 54$

$\therefore \frac{27x}{25} = 1080$

multiplying 25 on both side

$\therefore \frac{27x \times \cancel{25}}{\cancel{25}} = 1080 \times 25$

$\frac{\cancel{27}x}{\cancel{27}} = \frac{27000}{27}$

$\boxed{x = 1000}$

The cost the bicycle to Raju purchased is ₹ 1000

9. A Cricket player scored 180 runs in the first match and 257 runs in the second match. Find the numbers of run he should score in the third match so that the average of runs in the three matches be 230.

→ Suppose creckets score x in third match.

Total score by the cricketer $= 180 + 257 + x$

$\therefore$ Avarge run in match $= \frac{\text{Total Run}}{\text{No. of match}}$

$230 = \frac{180 + 257 + x}{3}$

$230 \times 3 = 437 + x$

$690 = 437 + x$

$x = 690 - 437$

$\boxed{x = 253}$

The cricketer should scored 253 runs in third match

10. Sudhir's present age is 5 more than three times the age of viru. Anil's age is half the age of Sudir. If

the ratio of the sum of sudhir's and Viru's age to three times Anil's age is 5:6, then find viru age.

→ Suppose. The age of viru be x years.

Then the age of sudhir is $(3x+5)$

Anil's age is half age is $\frac{3x+5}{2}$ years.

Sum of sudhir's and viru's age $= 3x+5+x$

$= 4x+5$

Three times anil's age $= 3\left[\frac{3x+5}{2}\right]$ years.

$\frac{4x+5}{3\left[\frac{3x+5}{2}\right]} = \frac{5}{6}$ (cross multiplication)

$6(4x+5x = 5 \mid 3)\left[\frac{3x+5}{2}\right]$

Multiplying both side 2

$2\times6(4x+5x = 5 \mid 3)\left[\frac{3x+5}{\cancel{2}}\right]\cancel{\times2}$

$12(4x+5x) = 15(3x+5)$

Divide both side by 3

$\frac{\cancel{12}^{4}(4x+5)}{\cancel{3}} = \frac{\cancel{15}^{5}(3x+15)}{\cancel{3}}$

$16x+20 = 15x+25$

$16x-15x = 25-20$

$\boxed{x = 5}$

The viru's age is 5 years.